

HW3: Instance Segmentation Report

Visual Recognition using Deep Learning, NYCU

朱驛庭, 111550093

1. Introduction

This work tackles the HW3 colored medical image instance segmentation task. The dataset contains 209 training images and 101 test images. Each image may contain four cell categories, and each non-zero connected instance id in `class1.tif-class4.tif` is an object mask. The evaluation metric is AP_{50} for instance segmentation, so the most important factors are recall, classification correctness, and avoiding duplicate instances; extremely precise mask boundaries are less critical than in $AP@[.50:.95]$.

The dataset is small but highly dense and imbalanced. The two frequent classes contain 14,537 and 15,653 instances, while class3 and class4 contain only 630 and 587 instances. Some images contain more than 700 cells, and image sizes range from tiny crops to images over 2000 pixels. These properties led us to use a Mask R-CNN-family method with small anchors, a high detection cap, multi-scale augmentation, and memory-safe streaming test-time augmentation (TTA).

The final best public submission is **0.5624 AP_{50}** , obtained by a TorchVision Mask R-CNN ResNet-50-FPN model with vertical/horizontal flip TTA and multi-scale mask fusion. This corresponds to public leaderboard rank 25 under team name **chueating**. The project repository is available at <https://github.com/ChuEating1005/VRDL/tree/main/HW3>.

2. Method

2.1. Model

We use Mask R-CNN [2] with a ResNet-50 [3] backbone and Feature Pyramid Network (FPN) [4]. ImageNet pretrained weights are used for the backbone only, satisfying the homework constraint. The model has **43.72M trainable parameters**, below the 200M parameter limit.

The key components are as follows. The **backbone** is ResNet-50, which extracts convolutional features at multiple spatial resolutions. The **neck** is FPN, which

builds a top-down feature pyramid so that small cells can be detected from high-resolution features while larger cells still use semantically strong deep features. The **RPN head** proposes candidate boxes on every FPN level using small anchors. The **RoI box head** classifies proposals into four cell classes and refines their bounding boxes. The **mask head** predicts a binary mask for each positive RoI with the standard BCE mask loss. The final submission converts the per-instance mask logits into binary masks and then COCO RLE.

The baseline configuration is designed for dense small objects:

- small anchors, with pyramid anchor sizes starting at 16 pixels;
- `detections_per_img=1000`, because a single test image can contain hundreds of cells;
- low box score threshold 0.05 to preserve recall for AP_{50} ;
- `model-side` `resize` `cap` `min_size=512`, `max_size=1024` to avoid dense-mask OOM.

We also implemented several ablation switches: ResNet-101, ConvNeXt-Tiny [6], PAFPN/PANet-style bottom-up path aggregation [5], BCE+Dice mask loss, Copy-Paste [1], and rare-class weighted sampling.

These modifications were not only hyper-parameter tuning. They change model components or training data distribution: PAFPN changes the neck topology, Dice changes the mask optimization objective, Copy-Paste changes the instance-level data distribution, rare-class sampling changes image sampling frequency, and 5-fold ensembling changes the inference model set.

2.2. Data Processing and Augmentation

TIFF images are loaded as RGB by dropping the constant alpha channel. For masks, each unique non-zero value is treated as one instance. Training augmentations include longest-side resize to 1024, random

scale in $[0.6, 1.6]$, another longest-side cap, horizontal/vertical flips, random 90-degree rotations, brightness/contrast, HSV jitter, CLAHE, and light blur.

Because class3 and class4 are rare, we tested two imbalance strategies. The first is binary rare-class weighted sampling: images containing class3 or class4 receive sample weight $1 + t$ with $t = 1.0$. The second is rare-class Copy-Paste, where class3/class4 instances are pasted onto the same canvas with probability 0.3. Both are intuitive, but both hurt public score in our experiments (see Sec. 4).

2.3. Training

The main baseline uses SGD with momentum 0.9, learning rate 0.0025, weight decay 10^{-4} , batch size 2, and 20 epochs. The trainable backbone layers are set to 3, and the RPN train/test pre-NMS and post-NMS proposal caps are 1000/500. The RoI box batch size per image is 256. We later trained a 30-epoch version, which achieved better validation AP_{50} but lower public score, suggesting validation overfitting on the small split. We also implemented an AdamW [7] + cosine schedule baseline inspired by prior solutions, but it was not selected as the final submission path.

All runs are logged to Weights & Biases under project `vr1d1-hw3`. Validation uses COCO-style mask AP and additionally logs per-class AP_{50} .

2.4. Streaming TTA and Submission Format

The largest gain came from inference. We use flip TTA over identity, horizontal flip, vertical flip, and horizontal+vertical flip. The best run also uses multi-scale TTA with scales 0.83, 1.0, and 1.17. For each augmented prediction, boxes and masks are mapped back to original image coordinates.

Naively accumulating all augmented outputs on GPU causes OOM because TorchVision pastes predicted masks back to full image resolution. Therefore TTA is performed in a streaming manner: after each augmented forward pass, predictions are moved to CPU candidates, GPU tensors are deleted, and CUDA cache is cleared. The final fusion is done on CPU.

We tested two TTA fusion modes. NMS fusion keeps the highest-scoring box per class after box NMS. Mask fusion groups predictions by box IoU, averages decoded binary masks weighted by confidence, thresholds the averaged mask, and converts it back to COCO RLE. Final records use the required COCO schema: 1-indexed `category_id`, `bbox=[x,y,w,h]`, and RLE `counts` encoded as strings.

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Figure 1. Final public leaderboard entry. The best submission reaches 0.5624 AP_{50} and rank 25.

3. Experiments

3.1. Submission Results

Tab. 1 summarizes all public submissions and important local experiments. The best result is 0.5624 AP_{50} from the baseline checkpoint plus vflip multi-scale TTA.

3.2. Effect of TTA

The most important finding is that TTA dramatically improves AP_{50} . Starting from the baseline score 0.5134, adding vertical/horizontal flip TTA and NMS raises performance to 0.5539 (+4.05 percentage points). Adding multi-scale inference further raises performance to 0.5624. This is consistent with the dataset: cells appear at diverse sizes, and AP_{50} rewards recovering more true instances.

Mask fusion and NMS fusion are close at single scale: 0.5525 vs. 0.5539. Multi-scale mask fusion gives the final best result, but its downside is inference cost. The final TTA setting requires 12 forward passes per image and must be streamed to prevent VRAM overflow.

3.3. Validation/Public Gap

The 30-epoch baseline achieved higher validation AP_{50} (0.52799 vs. 0.49902 for the 20-epoch baseline) but worse public AP_{50} (0.5558 vs. 0.5624). This indicates that the random validation split is not fully representative of the public test distribution. Therefore, after this point, public submissions were treated as the primary selection signal, while validation was used only for sanity checking.

3.4. Fold and Per-Class Variance

The 5-fold experiment revealed a stronger issue than ordinary overall fold variance: the best fold differs by class. As shown in Fig. 3, class1 is best on fold2, class2 on fold3, class3 on fold0, and class4 on fold4. The final AP_{50} spread is large: roughly 0.14 for class1, 0.09 for class2, 0.17 for class3, and 0.28 for class4. This means each fold learns a different class balance, so a single validation fold is a noisy proxy for the public test set.

4. Negative Results and Analysis

Several theoretically reasonable changes did not improve the public score.

Run	Main change	Epoch	Inference	Public AP ₅₀
baseline-v1	R50-FPN Mask R-CNN	20	none	0.5134
baseline-v1-vhflip-nms	same ckpt	20	vhflip + NMS fusion	0.5539
baseline-v1-vhflip-mask	same ckpt	20	vhflip + mask fusion	0.5525
baseline-v1-vhflip-nms-ms	same ckpt	20	vhflip + scales 0.83/1.0/1.17 + NMS fusion	0.5624
baseline-v1-ep30-vhflip-nms-ms	longer training	30	vhflip + scales 0.83/1.0/1.17 + NMS fusion	0.5558
r50-pafpn-vhflip-nms	PAFPN neck	30	vhflip + NMS fusion	0.5356
r50-pafpn-vhflip-nms-ms	PAFPN neck	30	vhflip + scales 0.83/1.0/1.17 + NMS fusion	0.5480
r50-cp-vhflip-nms-ms	Copy-Paste	30	vhflip + scales 0.83/1.0/1.17 + NMS fusion	0.5545
r50-rfs-cp-vhflip-nms-ms	RFS + Copy-Paste	30	vhflip + scales 0.83/1.0/1.17 + NMS fusion	0.5370
r50-rfs-vhflip-nms-ms	RFS	30	vhflip + scales 0.83/1.0/1.17 + NMS fusion	0.5312
r50-cp-ep20-tta	Copy-Paste	20	vhflip + scales 0.83/1.0/1.17 + NMS fusion	0.5410
baseline-v1-tta-det300	lower detection	20	vhflip + scales 0.83/1.0/1.17 + NMS fusion	0.5356
baseline-v1-ep20-fold0-tta	one CV fold	20	vhflip + scales 0.83/1.0/1.17 + mask fusion	0.4573
baseline-v1-ep20-5fold-tta	fused per-fold JSONs	20	post-hoc JSON mask fusion	0.5078
baseline-v1-ep20-5ckpt-direct-tta	direct 5-ckpt ensemble	20	model×TTA raw fusion	0.5042

Table 1. Public leaderboard AP₅₀ summary. The strongest improvement is TTA, not a heavier architecture.

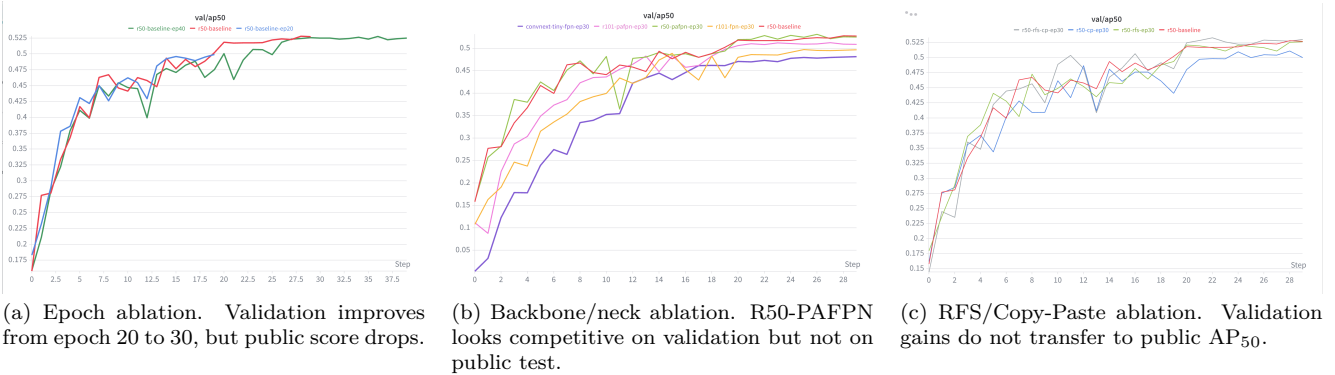


Figure 2. Representative Weights & Biases validation curves. These curves explain why validation-only model selection was unreliable for this small dataset.

For each additional experiment, we explicitly tracked the hypothesis, expected mechanism, result, and implication, as summarized in Tab. 2. This was useful because several ideas improved or looked competitive on validation but failed on the public leaderboard.

PAFPN. PAFPN adds a bottom-up path to enrich localization features. However, R50-PAFPN scored 0.5480 with multi-scale TTA, below the R50-FPN baseline. The likely reason is that the dataset has only 209 training images, and the extra neck capacity overfits or changes the feature distribution without improving AP₅₀ recall.

Rare-class sampling. Rare-class weighted sampling was expected to improve class3/class4 recall, but it scored 0.5312. Our implementation is intentionally simple: images containing class3 or class4 are sampled more frequently. This oversamples not only rare cells but also their backgrounds and co-occurring dense class1/class2 contexts, shifting the image distribution

away from test data.

Copy-Paste. Copy-Paste was less harmful than rare-class sampling but still below baseline: 0.5545 at epoch 30 and 0.5410 at epoch 20. It may increase rare-class variety, but the pasted instances can introduce unnatural overlaps and context mismatch. Since public AP₅₀ is dominated by dense class1/class2 recall, rare-class gains did not compensate for this distribution shift.

5-fold ensembling. We trained five baseline-v1 20-epoch folds and tried two ensemble strategies. Post-hoc fusion of already-fused per-fold JSONs scored 0.5078, and direct 5-checkpoint model×TTA fusion scored 0.5042. This shows that ensembling cannot rescue weak individual fold checkpoints. Since fold0 alone scored only 0.4573, the average model quality was much lower than the original baseline checkpoint. The per-class curves also explain why the ensemble was unstable: each fold specialized in different classes, but none was uniformly strong across all four classes. The direct

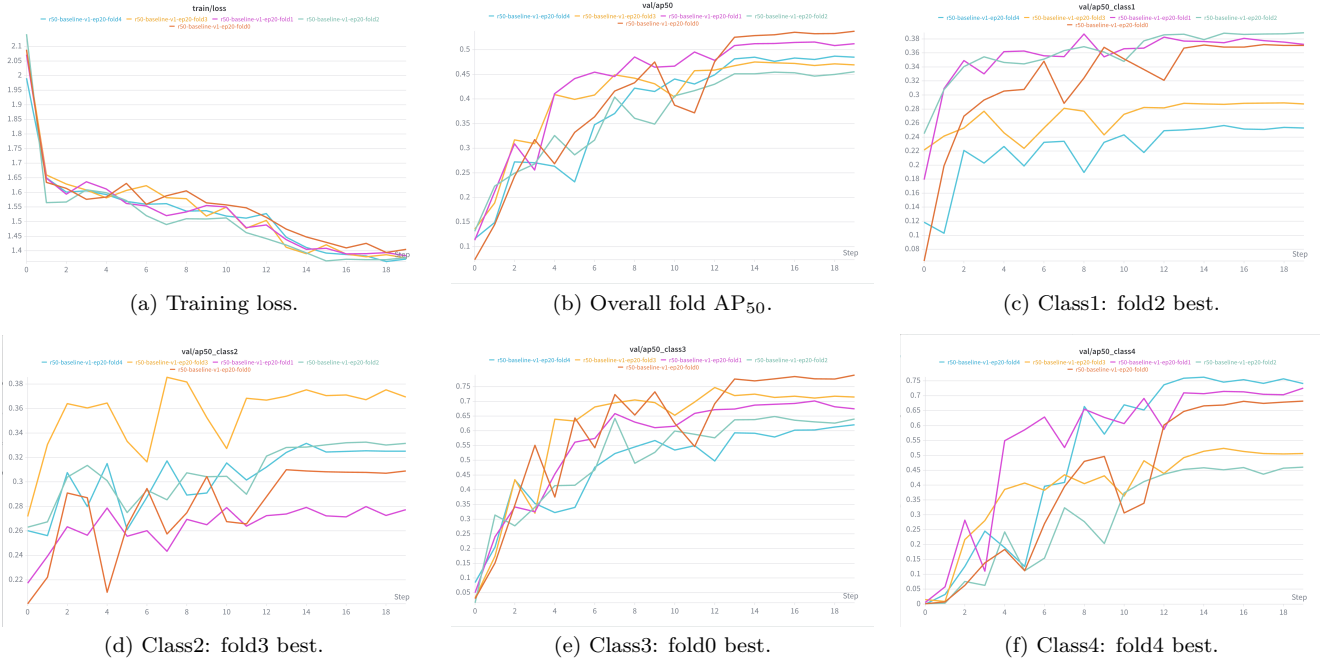


Figure 3. 5-fold validation curves. Different folds dominate different classes, exposing severe class-wise split sensitivity.

Experiment	Hypothesis	Mechanism	Result	Implication
Multi-scale TTA	flip Test cells have large scale and orientation variation.	Average predictions across flips/scales and fuse masks.	0.5134 → 0.5624 .	Strongest contribution; improves recall without retraining.
PAFPN neck	Bottom-up path aggregation should improve localization.	Add PANet-style feature aggregation to FPN.	0.5480, below baseline TTA.	Extra neck capacity did not transfer on 209 images.
Rare-class sampling	More class3/class4 exposure should improve rare-class AP.	Oversample images containing class3 or class4.	0.5312.	Image-level oversampling caused distribution shift.
Copy-Paste	Synthetic rare instances should improve instance diversity.	Paste class3/class4 masks into training images.	0.5545 at epoch 30; below best baseline.	Rare-class diversity was not enough to offset context mismatch.
5-fold ensemble	Fold diversity should reduce variance and improve robustness.	Train 5 stratified folds and fuse model/TTA candidates.	0.5042 direct ensemble.	Weak individual folds and class-specific fold bias hurt ensemble quality.

Table 2. Additional experiments beyond hyper-parameter tuning, with hypothesis, mechanism, result, and implication.

ensemble implementation is conceptually better than JSON-level double fusion, but the input models were not strong enough.

5. Conclusion

The final system uses TorchVision Mask R-CNN with ResNet-50-FPN, small anchors, high detection cap, strong geometric/color augmentation, and streaming multi-scale flip TTA. The best public AP₅₀ is **0.5624**, well above the strong baseline threshold. The main lesson is that, for this small dense-cell dataset, robust inference-time augmentation and memory-safe mask fusion were more valuable than heavier necks, rare-class oversampling, or naive k-fold ensembling.

GitHub repository: [https://github.com/](https://github.com/ChuEating1005/VRDL/tree/main/HW3)

[ChuEating1005/VRDL/tree/main/HW3](https://github.com/ChuEating1005/VRDL/tree/main/HW3).

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